

1. (c)

Given equation $x^{\log_x(2+x)^2} = 25 \Rightarrow (2+x)^2 = 25$ hold for $x = 3$

2. (b)

$$\begin{aligned} & \frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \frac{8}{7!} + \dots \infty \\ &= \frac{(1+1)}{1!} + \frac{(1+3)}{3!} + \frac{(1+5)}{5!} + \frac{(1+7)}{7!} + \dots \infty \\ &= \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \infty \right) + \left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \infty \right) \\ &= \frac{e - e^{-1}}{2} + \frac{e + e^{-1}}{2} = e \end{aligned}$$

3. (d)

$$\begin{aligned} T_n &= \frac{(2n+1)-1}{(2n+1)!} = \frac{1}{(2n)!} - \frac{1}{(2n+1)!} \\ \text{Here } S &= \sum_{n=1}^{\infty} T_n = \left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \infty \right) - \left(\frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \infty \right) \\ &\Rightarrow S = \left(\frac{e + e^{-1}}{2} - 1 \right) - \left(\frac{e - e^{-1}}{2} - 1 \right) \Rightarrow e^{-1} = \frac{1}{e} \end{aligned}$$

4. (d)

$$\begin{aligned} S &= \frac{1^3}{1!} + \frac{2^3}{2!} + \frac{3^3}{3!} + \dots + \frac{n^3}{n!} + \dots \\ \text{Here } T_n &= \frac{n^3}{n!} \Rightarrow \\ S_n &= \sum_{n=1}^{\infty} \frac{n^3}{n!} = 5e \end{aligned}$$

5. (d)

$$\begin{aligned} \frac{e^{7x} + e^x}{e^{3x}} &= e^{4x} + e^{-2x} = \sum_{n=0}^{\infty} \frac{(4x)^n}{n!} + \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!} \\ \text{We have } \frac{e^{7x} + e^x}{e^{3x}} &= \frac{4^n + (-2)^n}{n!} \\ \therefore \text{coefficient of } x^n \text{ in } \frac{e^{7x} + e^x}{e^{3x}} &= \frac{4^n + (-2)^n}{n!} \end{aligned}$$

6. (a)

$$\begin{aligned} \log_e 2 &= \frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \dots \infty \quad \dots\dots(i) \\ \text{We know that, } \log_e 2 &= 1 - \left(\frac{1}{2.3} \right) - \left(\frac{1}{4.5} \right) - \left(\frac{1}{6.7} \right) - \dots \infty \\ \text{Also } \log_e 2 &= 1 - \left(\frac{1}{2.3} \right) - \left(\frac{1}{4.5} \right) - \left(\frac{1}{6.7} \right) - \dots \infty \quad \dots\dots(ii) \end{aligned}$$

By adding (i) and (ii), we get,

$$\begin{aligned} 2 \log_e 2 &= 1 + \left(\frac{1}{1.2} - \frac{1}{2.3} \right) + \left(\frac{1}{3.4} - \frac{1}{4.5} \right) + \dots \\ \Rightarrow 2 \log_e 2 - 1 &= \frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - \frac{1}{4.5} + \dots \\ \Rightarrow \frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - \frac{1}{4.5} + \dots \infty &= \log_e 4 - \log_e e = \log_e \left(\frac{4}{e} \right) \end{aligned}$$

7. (b)

$$\begin{aligned} \text{We have, } \log_e(1+3x+2x^2) &= \log_e(1+x) + \log_e(1+2x) \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(2x)^n}{n} \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{1}{n} + \frac{2^n}{n} \right) x^n = \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{1+2^n}{n} \right) x^n \end{aligned}$$

$$x^n = (-1)^{n-1} \left(\frac{2^n + 1}{n} \right) = \frac{(-1)^{n+1}(2^n + 1)}{n}$$

So coefficient of

$$\left[\because (-1)^n = (-1)^{n+2} = \dots \right]$$

8. (c)

$$\begin{aligned} y &= \left[x^3 + \frac{(x^3)^2}{2} + \frac{(x^3)^3}{3} + \dots \right] = \log_e(1-x^3) \Rightarrow e^y = 1-x^3 \\ \Rightarrow x &= (1-e^y)^{\frac{1}{3}} \end{aligned}$$

9. (d)

$$\begin{aligned} T_n &= \frac{\sum_{n=1}^n n}{n!} = \frac{n(n+1)}{2 \cdot n!} \\ &= \frac{1}{2} \left[\frac{(n+1)}{(n-1)!} \right] = \frac{1}{2} \left[\frac{n-1}{(n-1)!} + \frac{2}{(n-1)!} \right] \\ &= \frac{1}{2} \left[\frac{1}{(n-2)!} + \frac{2}{(n-1)!} \right] \\ S_n &= \sum_{n=1}^{\infty} T_n = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{(n-2)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = \frac{e}{2} + e = \frac{3e}{2} \end{aligned}$$

10. (a)

$$\begin{aligned} x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty &= \log_e(1+x) \\ \text{We know that, } & \\ \text{Putting } x = 0.5, \text{ we get, } & \\ 0.5 - \frac{(0.5)^2}{2} + \frac{(0.5)^3}{3} - \frac{(0.5)^4}{4} + \dots \infty &= \log_e(1+0.5) = \log_e \left(\frac{3}{2} \right) \end{aligned}$$